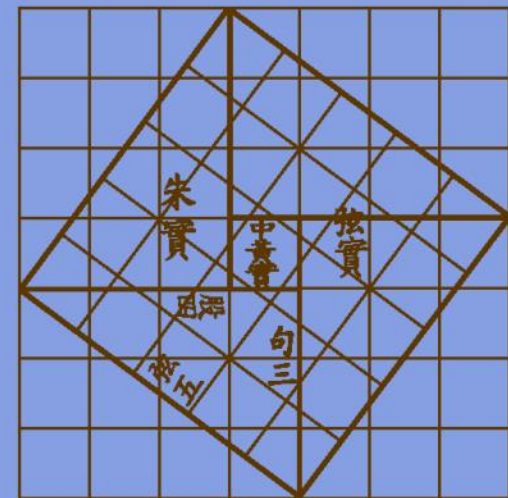
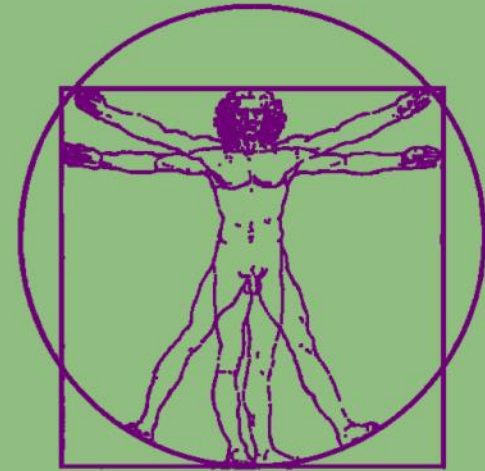
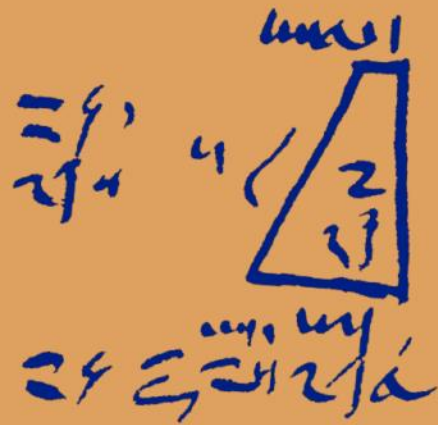


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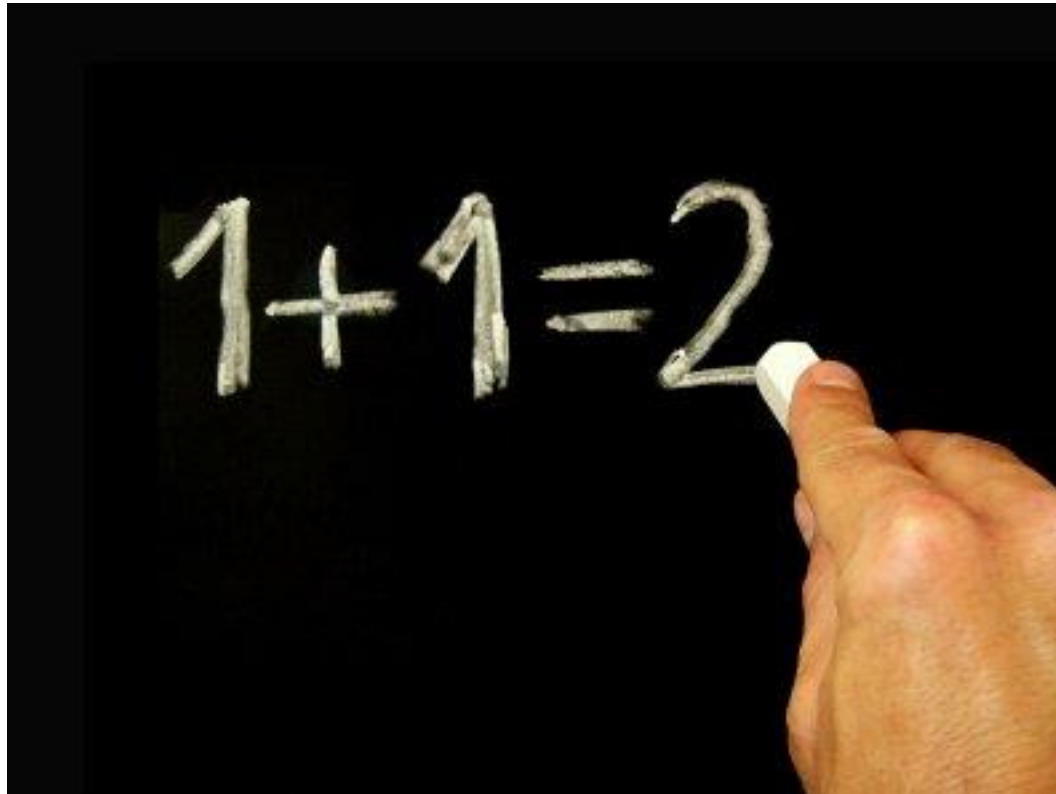
Mathematics: A Cultural Heritage

Lecture 7 – Part 2

Mathematics and Truth Some Philosophical Aspects of Mathematics



Mathematics = Truth?

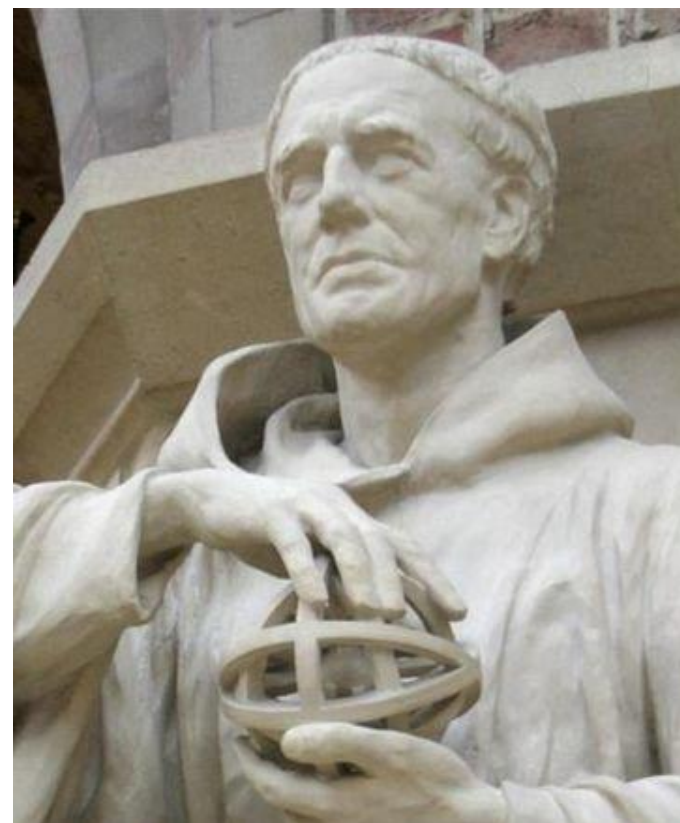




Quotes on Math & Truth

*“... if in other sciences we should arrive at **certainty without doubt and truth without error**, it behooves us to **place the foundations of knowledge in mathematics**, in so far as disposed through it we are able to reach certainty in other sciences and **truth by the exclusion of error.**”*

— Roger Bacon (English philosopher)



Roger Bacon (1214-1294)
Source: wikipedia



Quotes on Math & Truth

*“The most distinct and beautiful statement of any **truth** [in science] must take at last the **mathematical form.**”*

— Henry Thoreau (American writer and philosopher)



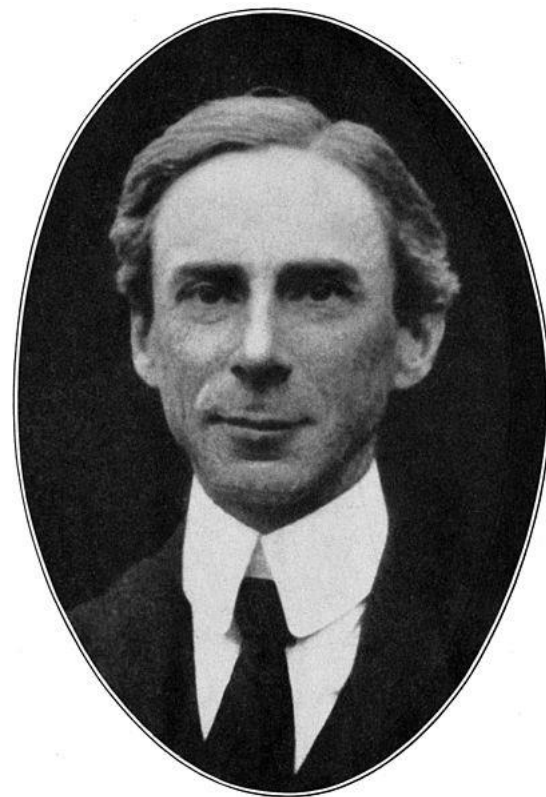
Henry Thoreau (1817-1862)
Source: wikipedia



Quotes on Math & Truth

*“Of these austerer virtues the love of truth is the chief, and **in mathematics, more than elsewhere, the love of truth** may find encouragement for waning faith. Every great study is not only an end in itself, but also a means of creating and sustaining a lofty habit of mind; and this purpose should be kept always in view throughout the teaching and learning of mathematics.”*

— Bertrand Russell



Bertrand Russell (1872-1970)

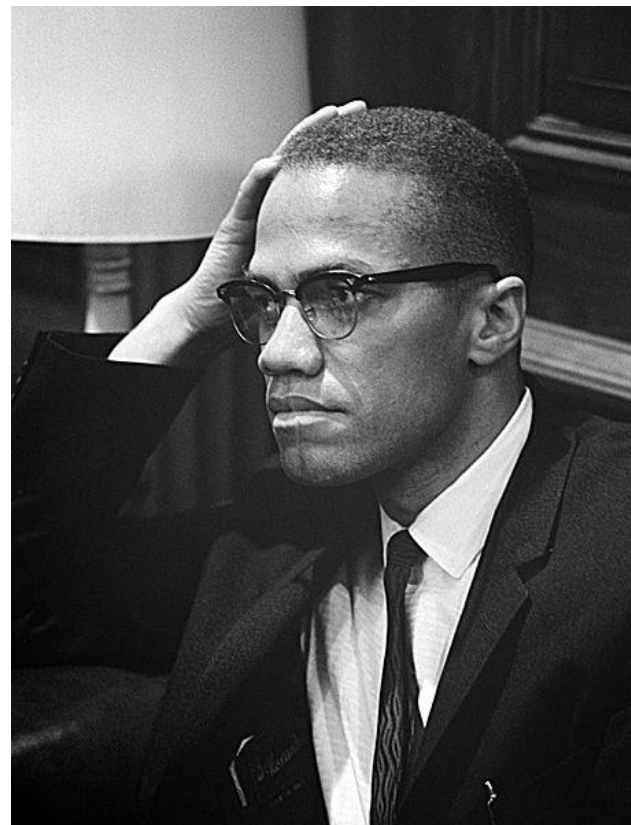
Source: wikipedia



Quotes on Math & Truth

*"I'm sorry to say that the subject I most disliked was mathematics. I have thought about it. I think the reason was that **mathematics leaves no room for argument**. If you made a mistake, that was all there was to it."*

— Malcolm X (American activist)



Malcolm X (1925-1965)

Source: wikipedia



Historical Examples of Mathematical Truths

- Ancient Greeks once believed that all quantities are **commensurable**:

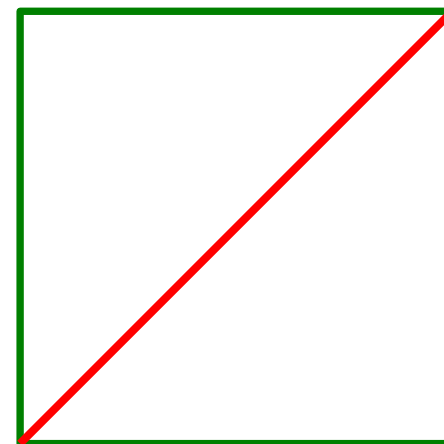
Given any two quantities (which can be represented as lengths of two line segments, or equivalently, two positive real numbers in today's terminology), there always exist a third quantity such that each of the two given quantities is a whole number multiple of the third quantity.



Commensurable Quantities



- In effect, this means all numbers are rational, which is not true!
- Recall that Hippiasus, a member of the Pythagorean school, proved that **the diagonal and the side of a square are not commensurable!**





Platonism/Realism

Mathematical Realist/Platonist view:

- Mathematical objects are abstract and exist by themselves, outside of space and time, independent of the human mind.
- Hence mathematical truths are:
 - **Absolute**
 - **Discovered, not invented**



Plato (c.428BC-348BC)



Other Opposing Views to Platonism

Logicism:

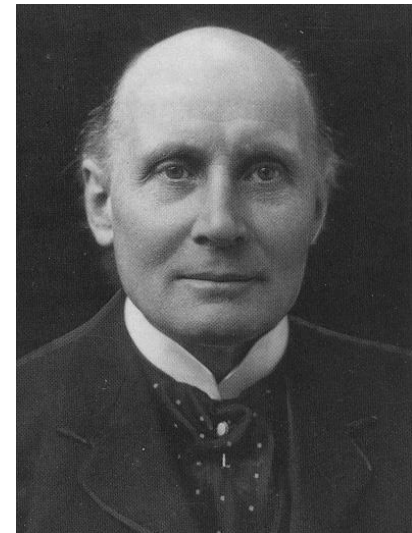
- Mathematics is an extension of **logic**
- Proponents: Gottlob Frege, Burtrand Russell, Alfred North Whitehead



Frege (1848-1925)



Russell

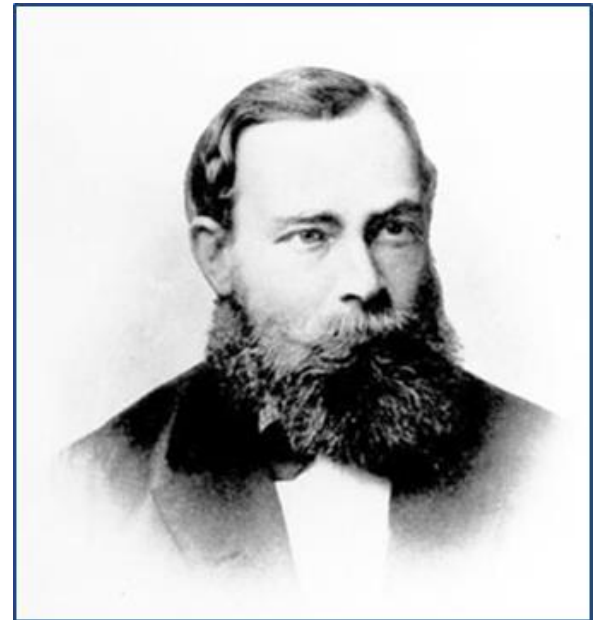


Whitehead (1861-1947)



Gottlob Frege

- Axiomatic predicate logic
- Gave formal meaning to statements
- Example: “Every boy likes some girl” vs. “Some girl is liked by every boy”.
- Repeat: Every boy likes some girl who likes some boy



Friedrich Ludwig Gottlob Frege
1848-1925



Set Theory

Set Theory

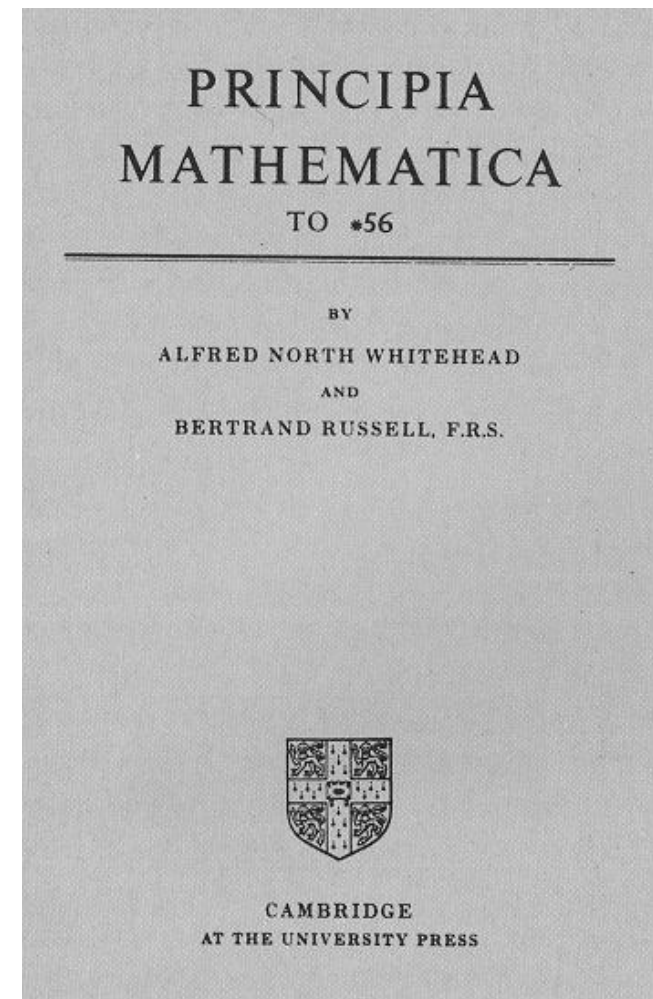
- A branch of mathematics which studies sets (collections) of objects.
- First developed by Georg Cantor and Richard Dedekind in late 1800s.
- Later developed by Ernst Zermelo, Abraham Fraenkel and others.
- Viewed as the most basic mathematical entities.
- Language of set theory can be used to describe all mathematics.



Math by Logic

Principia Mathematica

- Authored by Whitehead and Russell between 1910 and 1913.
- Three-volume monumental work aimed at providing a firm foundation of mathematics based on **logic** and **set theory**.



*54·42. $\vdash :: \alpha \in 2 . \supset :: \beta \subset \alpha . \mathfrak{A} ! \beta . \beta \neq \alpha . \equiv . \beta \in \iota''\alpha$

Dem.

$\vdash . *54·4 . \supset \vdash :: \alpha = \iota'x \cup \iota'y . \supset ::$

$\beta \subset \alpha . \mathfrak{A} ! \beta . \equiv :: \beta = \Lambda . \vee . \beta = \iota'x . \vee . \beta = \iota'y . \vee . \beta = \alpha : \mathfrak{A} ! \beta :$

[*24·53·56.*51·161] $\equiv :: \beta = \iota'x . \vee . \beta = \iota'y . \vee . \beta = \alpha$ (1)

$\vdash . *54·25 . \text{Transp.} . *52·22 . \supset \vdash : x \neq y . \supset . \iota'x \cup \iota'y \neq \iota'x . \iota'x \cup \iota'y \neq \iota'y :$

[*13·12] $\supset \vdash : \alpha = \iota'x \cup \iota'y . x \neq y . \supset . \alpha \neq \iota'x . \alpha \neq \iota'y$ (2)

$\vdash . (1) . (2) . \supset \vdash :: \alpha = \iota'x \cup \iota'y . x \neq y . \supset ::$

$\beta \subset \alpha . \mathfrak{A} ! \beta . \beta \neq \alpha . \equiv :: \beta = \iota'x . \vee . \beta = \iota'y :$

[*51·235] $\equiv :: (\mathfrak{A}z) . z \in \alpha . \beta = \iota'z :$

[*37·6] $\equiv :: \beta \in \iota''\alpha$ (3)

$\vdash . (3) . *11·11·35 . *54·101 . \supset \vdash . \text{Prop}$

*54·43. $\vdash :: \alpha , \beta \in 1 . \supset : \alpha \cap \beta = \Lambda . \equiv . \alpha \cup \beta \in 2$

Dem.

$\vdash . *54·26 . \supset \vdash :: \alpha = \iota'x . \beta = \iota'y . \supset : \alpha \cup \beta \in 2 . \equiv . x \neq y .$

[*51·231] $\equiv :: \iota'x \cap \iota'y = \Lambda .$

[*13·12] $\equiv . \alpha \cap \beta = \Lambda$ (1)

$\vdash . (1) . *11·11·35 . \supset$

$\vdash :: (\mathfrak{A}x, y) . \alpha = \iota'x . \beta = \iota'y . \supset : \alpha \cup \beta \in 2 . \equiv . \alpha \cap \beta = \Lambda$ (2)

$\vdash . (2) . *11·54 . *52·1 . \supset \vdash . \text{Prop}$

From this proposition it will follow, when arithmetical addition has been defined, that $1 + 1 = 2$.

*54·44. $\vdash :: z, w \in \iota'x \cup \iota'y . \supset_{z, w} . \phi(z, w) : \equiv . \phi(x, x) . \phi(x, y) . \phi(y, x) . \phi(y, y)$

Dem.

$\vdash . *51·234 . *11·62 . \supset \vdash :: z, w \in \iota'x \cup \iota'y . \supset_{z, w} . \phi(z, w) : \equiv :$

$z \in \iota'x \cup \iota'y . \supset_z . \phi(z, x) . \phi(z, y) :$

[*51·234.*10·29] $\equiv : \phi(x, x) . \phi(x, y) . \phi(y, x) . \phi(y, y) : . \supset \vdash . \text{Prop}$

*54·441. $\vdash :: z, w \in \iota'x \cup \iota'y . z \neq w . \supset_{z, w} . \phi(z, w) : \equiv :: x = y : \vee : \phi(x, y) . \phi(y, x)$

Dem.

$\vdash . *5·6 . \supset \vdash :: z, w \in \iota'x \cup \iota'y . z \neq w . \supset_{z, w} . \phi(z, w) : \equiv ::$

$z, w \in \iota'x \cup \iota'y . \supset_{z, w} : z = w . \vee . \phi(z, w) : .$

[*54·44] $\equiv :: x = x . \vee . \phi(x, x) : x = y . \vee . \phi(x, y) :$

$y = x . \vee . \phi(y, x) : y = y . \vee . \phi(y, y) :$

Using this approach, It took the authors 379 pages to prove that $1+1=2$.



Russell's Paradox

- Discovered by Russell in 1901.
- Shattered the Foundation of (Naïve) Set Theory of Cantor.

The Paradox:

- Suppose any collection of definable objects will form a set.
- Suppose the collection of all things (including sets) is also a set (which is the **universal set**).
- Usually a set is not a member of itself (e.g, the set of all triangles); but some set can be a member of itself (e.g., the set that contains all things which are not triangles).



Russell's Paradox

The Paradox (continued):

- Now define

R = the set of all sets which are not members of itself

- Question: is R a member of itself?
 - If R is a member of itself, then ...
 - If R is not a member of itself, then ...
- In math language:

Let $R = \{x \mid x \notin x\}$. Then $R \in R \Leftrightarrow R \notin R$ — a contradiction!

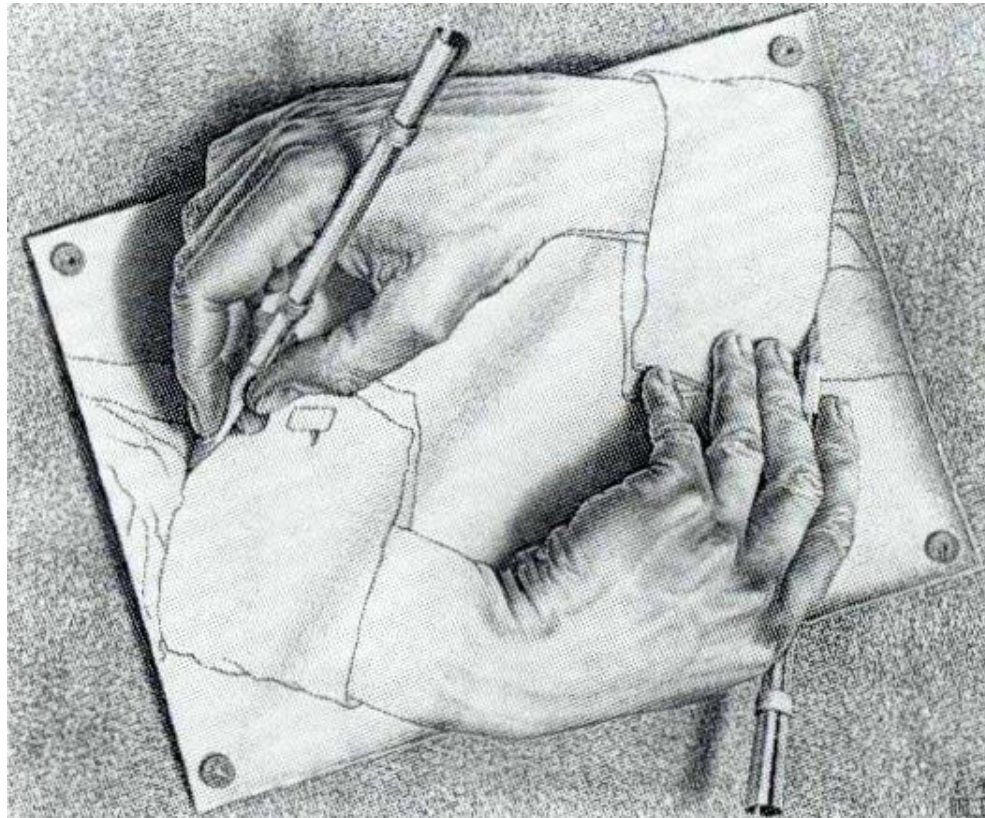


The Barber Paradox

- In a village there is only 1 (male) barber.
- Among the other (male) villagers, some shave themselves, and those who do not shave themselves will have the barber's service.
- The barber sets a rule: **he shaves only those who do not shave themselves.**
- **Question: Does the barber shave himself?**



The Strange Loop of Self-reference!





Fixing Russell's Paradox

Paradoxes need to be fixed!

- Proposed solution: The naïve ***R*** is not a set.
 - It is a “class”
 - The paradox is precisely that ***R*** cannot “live” anywhere
- Proposed solution: elements in sets must “live” somewhere
 - If we have a set ***S***, can define $T = \{x \in S \mid x \notin x\}$.
 - Then ***T*** is a set.
 - However, ***T*** isn't in ***S*** (there's a hierarchy).



Other Opposing Views to Platonism

Mathematical Formalism

- According to Davis and Hersh in their book *The Mathematical Experience* (1981):

*“Mathematics is the science of rigorous proof”... based on “some undefined terms, and some unproved statements about these terms.” (Such statements are called **axioms.**)”*

*“One cannot assert that a theorem is true, any more than one can assert that the axioms are true... Thus the statements of mathematical theorems have no content at all; they are not about anything. On the other hand, **according to the formalist, they are free of any possible doubt or error, because the process of rigorous proof and deduction leaves no gaps or loopholes.**”*



Mathematical Formalism

- Proponent: **David Hilbert**
- It was the prevailing philosophy around early-mid 1900's.



David Hilbert (1862-1943)

Source: wikipedia



Hilbert's Program

Hilbert proposed a **Program** in early 1920's. He envisaged:

- All mathematics can be **formalized** (using the formalist approach).
- A proof of **consistency** of mathematics: no mathematical statements can be true and not true at the same time.
- A proof of **completeness** of mathematics: all true mathematical statements can be proved in finitely many steps.
- An algorithm to decide whether any mathematical statement is true or false.



An Analogy

Newtonian Physics and the **Mechanical Worldview**



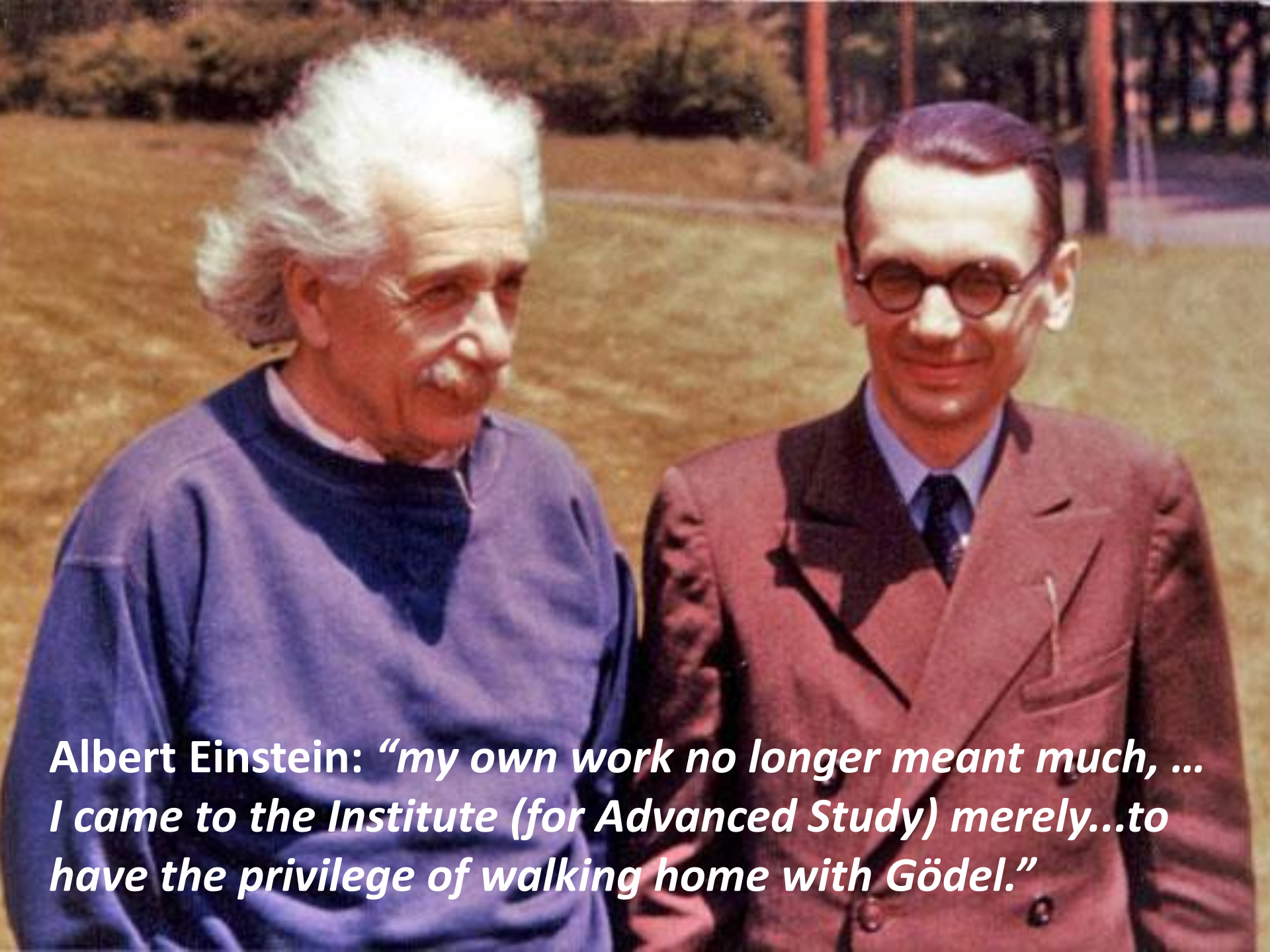
Gödel's Work

Kurt Gödel

- Logician and mathematician
- Born in Austria-Hungary, emigrated to USA after WWII.
- Most important contribution: the **Incompleteness Theorems** (1931).
- Also proved results about the **Axiom of Choice** and the **Continuum Hypothesis**.



Kurt Gödel (1906-1978)
Source: wikipedia



***Albert Einstein: “my own work no longer meant much, ...
I came to the Institute (for Advanced Study) merely...to
have the privilege of walking home with Gödel.”***



Gödel's Incompleteness Theorems

The First Incompleteness Theorem

In any axiomatic mathematical system which is

- Consistent (i.e., no statement in the system can be true and not true at the same time), and
- Capable of doing arithmetic of natural numbers (i.e., counting numbers),

there are always mathematical statements about natural numbers which are **true** but **unprovable** within the system.



Implication

- Any consistent axiomatic mathematical system cannot be complete!

Hilbert's program is doomed!



Gödel's Incompleteness Theorems

The Second Incompleteness Theorem

No axiomatic mathematical system can prove its consistency within itself.

(note: this in fact is a corollary of the First Incompleteness Theorem)

So, Hilbert's program is doomed again!



Gödel's Incompleteness Theorems



<https://www.youtube.com/watch?v=YrKLy4VN-7k>



Significance of Gödel's Results

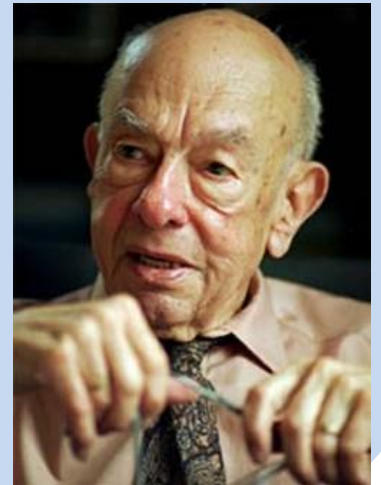
- Shows limitation of the axiomatic approach.
- Limitation of human's knowledge?
- End of rationalism?
- Comparison to other discoveries/theories around the same time:
 - **Uncertainty Principle in physics** (which states that it is impossible to measure **position** *and* **momentum** of objects precisely at the same time)
 - **Quantum Mechanics**
 - **Relativity**



Another uncertainty

- Statement might seem silly
- Halting problem (computer science):
Is there a computer program which tells you whether any other program will eventually stop?
- Answer: No! If such a program P exists, then write program: “If P says that I stop, then run forever, if P says that I run forever, then stop!”

“The totality of our so-called knowledge or beliefs, from the most casual matters of geography and history to the profoundest laws of atomic physics or even of pure mathematics and logic, is a man-made fabric which impinges on experience only along the edges. Or, to change the figure, total science is like a field of force whose boundary conditions are experience. A conflict with experience at the periphery occasions readjustments in the interior of the field. Truth values have to be redistributed over some of our statements. Reevaluation of some statements entails reevaluation of others, because of their logical interconnections—the logical laws being in turn simply certain further statements of the system, certain further elements of the field.”



— Willard Van Orman Quine (1908-2000, American philosopher)



General Perception of Math as Truth

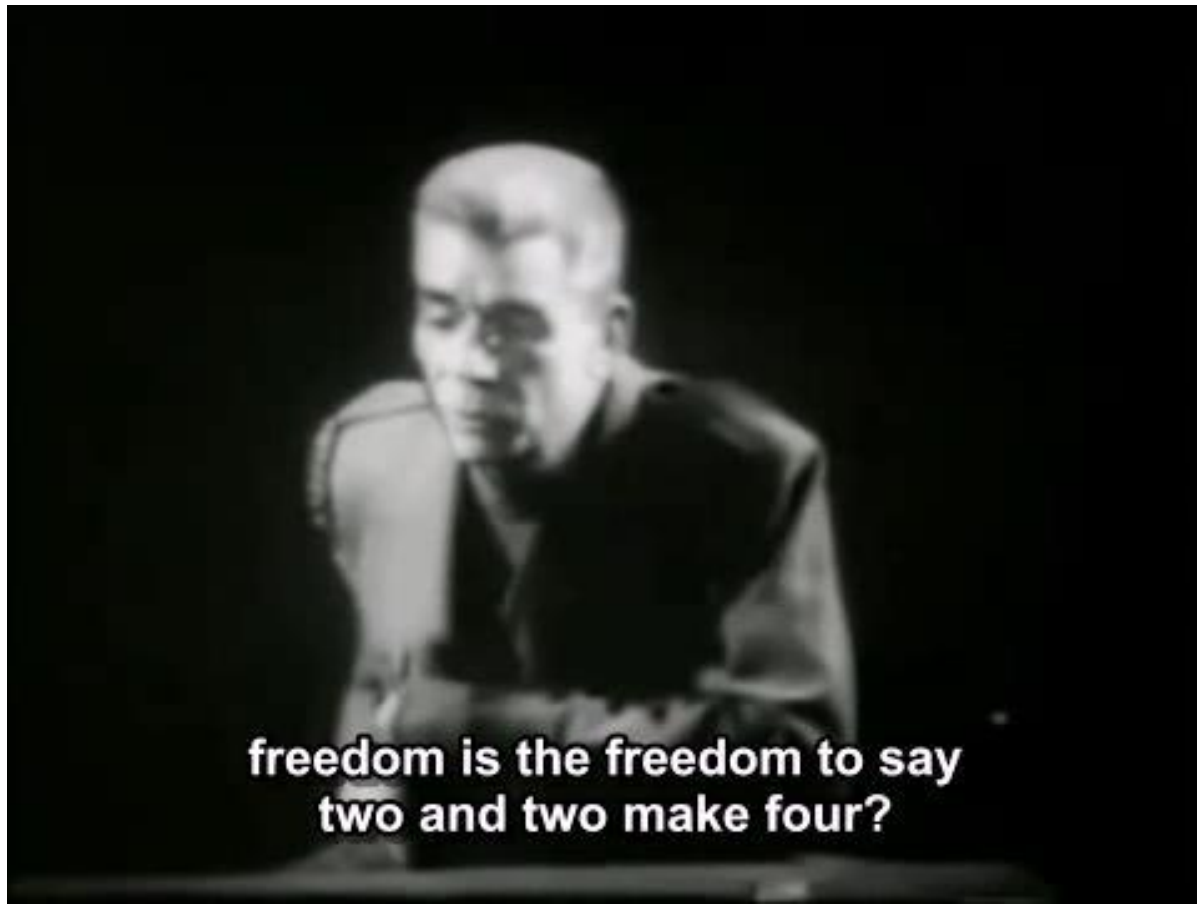
- Video: $2 + 2 = \dots$





General Perception of Math as Truth

- Video: $2 + 2 = \dots$ (in 1984)





Topic of Next Lecture

Theorems, Problems, and Proofs